



ARTIFICIAL INTELLIGENCE

These challenges are aggravating as computers become more pervasive and take part in our daily lives in many ways. It does not take a genius to tell that the computers do not have a brain of their own, let alone intelligence or conscience. It is the developer who instructs and teaches the computers what to do. If the developer makes a mistake, and designs and develops the code poorly, it can be a problem. If he or she does not test the development appropriately, use sub-par hardware, or if there are fundamental flaws in the understanding of user requirements, the computers will perform poorly.

So, what is the problem?

The problem is not the emerging technology and the bright future it brings with it. It is our adamant belief that there is a sunny technological future just around the corner!

Many businesses, and by that I mean senior managers in those businesses, still feel and believe that the Internet of Things (IoT), artificial intelligence (AI) or automation (RPA) will solve their problems for good, only to find themselves in a differently flavoured soup post-implementation. Why do you think this might be the case?

In my view, it starts with raising expectations early in the adoption process, when the person in charge of such initiatives demonstrates only one side of the story. He or she usually does not tell the other side, either because of lack of full knowledge or because of some special interest in doing so (often happens with vendors of technology). Accepting the fact that we do not know what we do not know is quite critical here.

This problem further increases with unrealistic expectations from the technology as well as failing to define the acceptance criteria upfront. Failing to define a criteria upfront only results in the endowment effect. Implementation teams try to justify (much later) that whatever has been developed should be consumed because they worked hard on it; worse part is when they try to retrofit the acceptance criteria only to make it happen.

So essentially, businesses do not have stubborn goals (read acceptance

criteria) and do not have a handle on the means. This results in a somewhat uncontrollable situation. Development teams may tell you that the machine will learn eventually, but they will not tell you when and how would it improve.

Garbage-in will always result in garbage-out, no matter how many years you keep doing it and how intelligent the machine is!

Know thy limits...

Any machine or AI cannot differentiate between right and wrong. It can only choose what is popular as it sees from learned data. There is a fundamental problem with AI, unfortunately; it learns from the data fed to it. Whether it is supervised or unsupervised, does not matter. Data must be good and balanced. If we want to teach the machine with examples, they must be good.

If for some reason that (clean data) cannot be ensured, then testing of developed AI must be flawless. If testing has gaps and data is bad, bad AI will rise. It will not only have garbage-in to give garbage-out, but it will also do it at a much faster rate and scale. No one wants that. And therefore we need to know these limitations and deal appropriately with emerging technologies.

There are several challenging aspects that an AI machine cannot handle. Virtues such as fairness, morality and ethics cannot be taught to the machine and, hence, the machine cannot make certain judgement calls based on those.

Many narrow AI programs are also not flawless per se. These programs merely try to imitate human behaviour (which itself is questionable sometimes). If choices are black and white, it works well, but soon it folds up when the problems go in the grey area. Poorly-designed programs then tend to make random (often wrong) choices, costing businesses heaps of money. Many businesses feel that this is perhaps an acceptable error-rate, which sometimes could be an unfounded assumption. Especially when acceptance criteria were not fixed before starting the journey.

However, from a larger picture

perspective, only the narrow AI is the lesser of two evils. Anything further would mean we have to define and codify a lot of grey matter—and humans have limits!

What are the takeaways?

For sure, there is a lot to talk about teaching machines morality, ethics, working in grey areas and many things alike. However, we cannot wait for all the lights to turn green, and we must keep moving forward, learn and improvise. However, the biggest takeaway, for now, would be to remain positively sceptical, keep the sensibility hat always on and adopt the technology with a pinch of salt.

Machines make mistakes, just like humans do, and they will keep making them in the future, too. Businesses must accept this fact and know that machines also need—much like humans—attention, retraining and performance improvement plan before they go live again.

Businesses must make sure—just as they do for humans—machines are progressively-trained and rigorously-tested before giving them more responsibilities. Any failure of machines' performance should be dealt with, rather relatively stronger than humans.

I also suggest that businesses should establish or augment their existing HR department to HAIR (human and artificial intelligent resources) department. Appropriate policies should be developed for managing those AI resources, just as they do for humans. This may sound a bit silly for now, but the direction we are heading towards would soon dictate that. A movement towards making AI transparent is catching up.

Lastly, do not get carried away and assume that just because we have a cool technology, we can use it to solve every problem around us. Emerging technologies are new hammers; let us avoid treating all the problems as nails and avoid rushing into the emerging future. It is extremely difficult to undo strategic and technological mistakes these days.

Sometimes, it is better to deal with humans than machines; sanity is the key! **EFY**